

Astrodynamics I: Midterm Exam

April 21, 2014 7:00-8:30PM (90 minutes)

You may use the following constants and formulae if necessary.

- Radius of Earth $R_E = 6.378 \times 10^3 \text{ km} = 2.09 \times 10^7 \text{ ft}$
- Gravity Parameter of Earth $\mu_E = 3.986 \times 10^5 \text{ km}^3 / \text{sec}^2 = 1.407 \times 10^{16} \text{ ft}^3 / \text{sec}^2$
- $\vec{e} = \frac{1}{\mu} \left[\left(v^2 - \frac{\mu}{r} \right) \vec{r} - (\vec{r} \cdot \vec{v}) \vec{v} \right]$, $e = \sqrt{1 + \frac{2Eh^2}{\mu_E^2}}$

- (20) A space object enters the atmosphere of the Earth at an altitude of 350,000ft with a velocity of 23,000ft/sec and a flight path angle of -30 degrees.
 - (6) What type of orbit does the space object follow?
 - (6) What were the velocity vector and flight path angle of the space object when it was at an altitude of 528,000ft?
 - (8) Ignoring the effect of the Earth's atmosphere on the motion of the object, determine if the space object impacts the surface of the Earth or not. If it does, what is its impact velocity vector? If not, what is its minimal miss altitude?
- (30) (f and g functions)

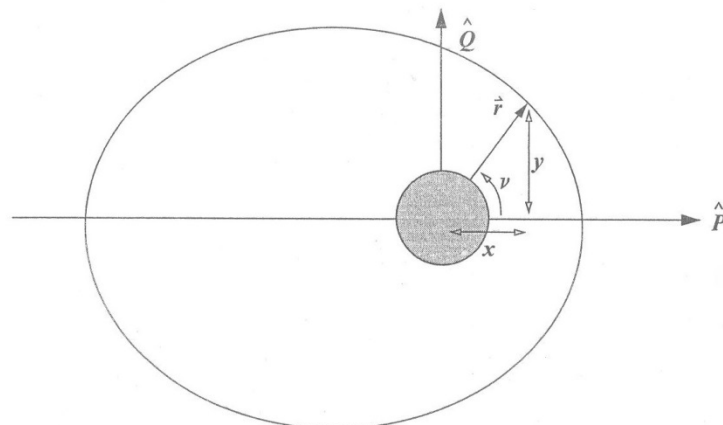


Figure 2-14. Geometry for the Perifocal Coordinate System. We can break a satellite's position into two components (x, y) in the orbital plane.

The position and velocity vectors can be expressed in perifocal coordinate system as

$$\begin{cases} \vec{r} = x\hat{P} + y\hat{Q} \\ \vec{v} = \dot{x}\hat{P} + \dot{y}\hat{Q} \end{cases}$$

It is assumed that the initial position and velocity vectors $(\vec{r}_0, \vec{v}_0)$ are known. Then, the position and velocity vectors at an arbitrary time can be also expressed as

$$\begin{cases} \vec{r} = f\vec{r}_0 + g\vec{v}_0 \\ \vec{v} = \dot{f}\vec{r}_0 + \dot{g}\vec{v}_0 \end{cases}$$

- a. (10) Derive $(f, \dot{f}, g, \dot{g})$ as functions of (x, y, x_0, y_0) and their derivatives.
 - b. (5) Discuss how to check the accuracy of (f, g) functions.
 - c. (15) Suppose that the true anomaly ν is known. Derive (f, g) as functions of the change in true anomaly $\Delta\nu = \nu - \nu_0$.
3. (20) Determine the orbital elements for an Earth orbit which has the following position and velocity vectors (set the gravitational parameter $\mu = 1$):
- $$\begin{cases} \vec{r} = 1K \text{ DU} \\ \vec{v} = 1I \text{ DU/TU} \end{cases}$$
4. (20) Clearly explain the procedure of how to obtain the state vectors $(\vec{r}, \vec{v})$ from the classical orbital elements $(a, e, i, \Omega, \omega, \nu)$.
5. (10) Explain Kepler's problem **succinctly** and **precisely**.